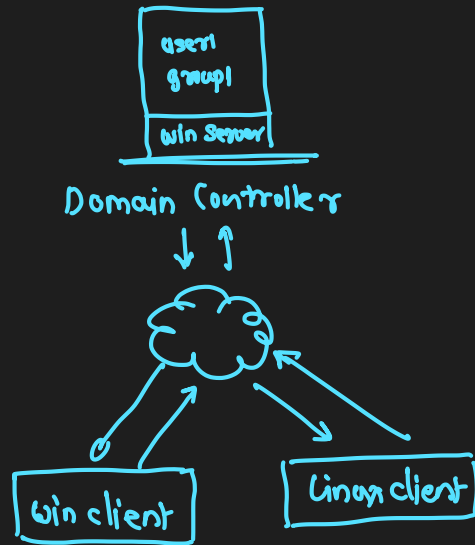




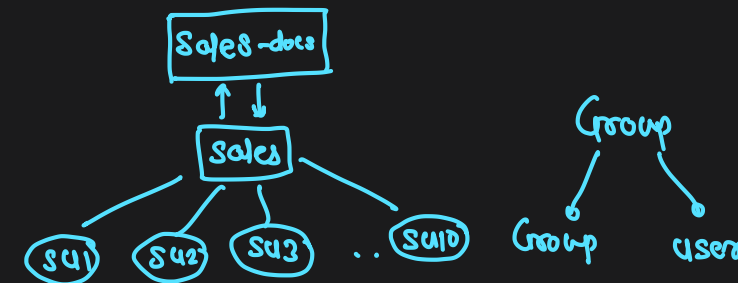
Users and Group Management

Introduction



- User and group management is the day-to-day operational core of Active Directory administration
- Every person who logs into the network, every computer that joins the domain, and every service that authenticates requires a user or computer account
- Groups then tie these accounts to permissions and policies
- **Core Objectives**
 - **Identity Management:** Create, modify, and delete user identities → authentication
 - **Access Control:** Determine who can access what resources → authorization → permissions | Group policy
 - **Administrative Efficiency:** Use groups to assign permissions at scale
 - **Security:** Enforce least privilege, monitor privileged accounts
 - **Lifecycle Management:** Handle onboarding, role changes, and offboarding

default security principle → implicit deny





Users



User object



- A user object represents a security principal that can:
 - Authenticate to the domain (logon) → authentication
 - Be assigned permissions to resources → authorization
 - Be a member of groups → group members
 - Own files and objects → owner
 - Have a mailbox (if Exchange is deployed)
- Types of users

Type	Purpose	Characteristics
✓ <u>Regular User</u> → human user	<u>Standard employee/end-user</u>	<u>Limited permissions; interactive logon</u>
✓ <u>Administrative User</u> → superusers (root)	<u>IT staff with elevated rights</u>	<u>Member of admin groups; separate admin account</u>
✓ <u>Service Account</u> → system user	<u>Used by applications/services</u>	<u>Non-interactive; often no password expiry</u>
<u>gMSA (Group Managed Service Account)</u>	<u>Modern service account</u>	<u>Automatic password management; ideal for services</u>
<u>Managed Service Account (MSA)</u>	<u>Single-server service account</u>	<u>Automatic password; limited to one server</u>
<u>Shared Account</u> → disabled	Multiple users (discouraged)	Hard to audit; security risk
<u>Guest Account</u> → disable	<u>Temporary/limited access</u>	<u>Disabled by default; high risk if enabled</u>

User Account Attributes



■ General Attributes

- First name, Last name, Display name
- Description, Office, Telephone, Email
- Web page, Notes

■ Account Attributes

↪ user principal name
↪ samuseraccount@domain name

- User logon name (UPN): user@domain.com
- User logon name (pre-Windows 2000): DOMAIN\user
- Logon hours
- Logon workstations
- Account options (password never expires, etc.)
- Account expiry date

■ Profile Attributes

- Profile path
- Logon script
- Home folder (local or network)
- Remote Desktop Services profile

■ Organization Attributes

- Title, Department, Company
- Manager
- Direct reports

■ Address Attributes

- Street, City, State, ZIP
- Country/region

■ Telephones Attributes

- Home, Pager, Mobile, Fax, IP phone

■ Member Of **

- Group memberships

■ Dial-in

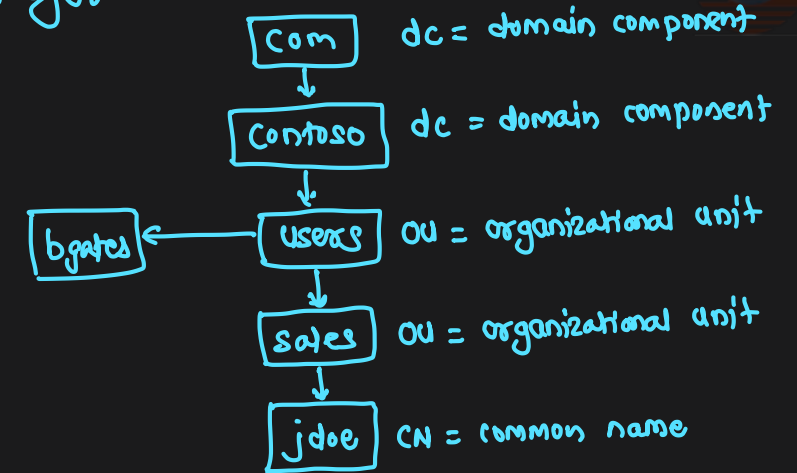
- Remote access permission
- Callback options
- Static IP address

Creating User Accounts

Basic user creation

```
New-ADUser -Name "John Doe" `
  -GivenName "John" ` → first name
  -Surname "Doe" ` → last name
  -SamAccountName "jdoe" ` → unique name used to login
  -UserPrincipalName "jdoe@contoso.com" `
  -Path "OU=Sales,OU=Users,DC=contoso,DC=com" `
  -AccountPassword (ConvertTo-SecureString "P@ssw0rd123!" -AsPlainText -Force) `
  -Enabled $true `
  -ChangePasswordAtLogon $true `
  -Department "Sales" `
  -Title "Sales Representative" `
  -Manager "cn=Jane Smith,ou=Managers,dc=contoso,dc=com"
```

SAM → Security Account Manager



Managing User Accounts



Find user ↗ Sam Account Name | SID | object GUID

Get-ADUser -Identity "jdoe" -Properties *

Find users by filter

Get-ADUser -Filter {Department -eq "Sales"} -Properties Department, Title

Search by name

Get-ADUser -Filter {Name -like "*John*"}

Reset password

Set-ADAccountPassword -Identity "jdoe" `
 -Reset `
 -NewPassword (ConvertTo-SecureString "NewP@ssw0rd!" -AsPlainText -Force)

Force password change at next logon

Set-ADUser -Identity "jdoe" -ChangePasswordAtLogon \$true

Unlock account

Unlock-ADAccount -Identity "jdoe"

Comparison operators

→ -eq → ==

→ -ne → !=

→ -gt → >

→ -lt → <

→ -ge → >=

→ -le → <=

→ -like → like

Managing User Accounts



Disable account

Disable-ADAccount -Identity "jdoe"

Enable account

Enable-ADAccount -Identity "jdoe"

Move user to different OU

Move-ADObject -Identity "CN=John Doe,OU=Sales,DC=contoso,DC=com" `

-TargetPath "OU=Marketing,DC=contoso,DC=com"

Modify attributes

Set-ADUser -Identity "jdoe" -Department "Marketing" -Title "Marketing Specialist"

Add to group

Add-ADGroupMember -Identity "Marketing Team" -Members "jdoe"

Remove from group

Remove-ADGroupMember -Identity "Sales Team" -Members "jdoe" -Confirm:\$false

Delete user

Remove-ADUser -Identity "jdoe" -Confirm:\$false

Bulk User Creation



- CSV format

- FirstName,LastName,SamAccountName,UPN,Department,Title,OU>Password
- John,Doe,jdoe,jdoe@contoso.com,Sales,Representative,"OU=Sales,DC=contoso,DC=com",P@ssw0rd123!
- Jane,Smith,jsmith,jsmith@contoso.com,Marketing,Specialist,"OU=Marketing,DC=contoso,DC=com",P@ssw0rd123!

```
Import-Csv "C:\NewUsers.csv" | ForEach-Object {  
    $params = @{  
        Name = "$($_.FirstName) $($_.LastName)"  
        GivenName = $_.FirstName  
        Surname = $_.LastName  
        SamAccountName = $_.SamAccountName  
        UserPrincipalName = $_.UPN  
        Path = $_.OU  
        Department = $_.Department  
        Title = $_.Title  
        AccountPassword = (ConvertTo-SecureString $_.Password -AsPlainText -Force)  
        Enabled = $true  
        ChangePasswordAtLogon = $true  
    }  
    New-ADUser @params  
    Write-Host "Created user: $($_.SamAccountName)"  
}
```

User Account Lifecycle Management



■ Onboarding

- Create user account (HR provides details)
- Assign to appropriate groups
- Set initial password; force change at first logon
- Configure mailbox (if Exchange)
- Set up home folder/profile
- Provide credentials to user securely

■ Offboarding

- Disable account immediately (don't delete)
- Remove from all groups (except Domain Users)
- Reset password to random value
- Hide from address lists (Exchange)
- Set out-of-office (Exchange)
- Forward email (if needed)
- Document and archive
- Delete after retention period (e.g., 30-90 days)

■ Role Changes

- Update group memberships (remove old, add new)
- Update attributes (department, title, manager)
- Review permissions
- Update mailbox/delegation if needed

■ Best Practices

- **Never delete immediately:** Disable first; delete after retention period
- **Use AD Recycle Bin:** Enable for object recovery
- **Document everything:** Who, what, when, why
- **Automate where possible:** PowerShell scripts, HR system integration
- **Review regularly:** Quarterly access reviews



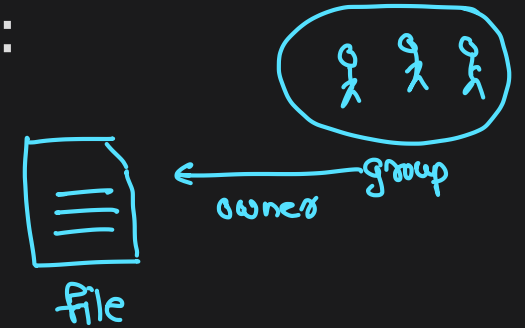
Groups

Group of security principles

What is a Group?



- A group is a collection of security principals (users, computers, groups) used to:
 - Simplify permission assignment
 - Simplify policy application
 - Enable email distribution (distribution groups)
 - Organize objects for management



- **Group Types**

→ files | folders | computers

Type	Purpose	Can be Used for Permissions?
<u>Security Group</u>	<u>Assign permissions to resources</u>	<u>Yes</u>
<u>Distribution Group</u>	<u>Email distribution lists</u>	<u>No (only email)</u>

Group Scopes



Scope	Can Contain	Can Be Used In
<u>Domain Local</u>	<u>Users, computers, global groups, universal groups from any domain</u>	<u>Permissions on resources in the same domain</u>
<u>Global</u>	<u>Users, computers from the same domain</u>	<u>Permissions on resources in any domain</u>
<u>Universal</u>	<u>Users, computers, global groups from any domain</u>	<u>Permissions on resources in any domain</u>

■ Group Scope Best Practices

■ AGDLP (Accounts, Global, Domain Local, Permissions)

- Accounts: Add users to Global groups (by role/department)
- Global groups: Add to Domain Local groups (by resource)
- Domain Local groups: Assign Permissions to resources

■ Example

- User: John Doe (Sales)
- Global Group: Sales Team (contains all sales users)
- Domain Local Group: Sales Shared Folder Access (contains Sales Team global group)
- Permission: Sales Shared Folder Access has Modify on \\server\sales



Built-in Domain Local Groups

Group	Purpose
<u>Administrators</u>	<u>Full control over the domain</u>
<u>Account Operators</u>	<u>Create/modify user and group accounts</u>
<u>Backup Operators</u>	<u>Backup/restore files and registry</u>
<u>Guests</u>	<u>Limited access (disabled by default)</u>
<u>Print Operators</u>	Manage printers
<u>Server Operators</u>	Manage servers
<u>Users</u>	<u>Standard user access</u>
Performance Monitor Users	Access performance data
Performance Log Users	Manage performance logs
Remote Desktop Users	Remote Desktop access
Network Configuration Operators	Manage network settings
Incoming Forest Trust Builders	Create incoming forest trusts
Terminal Server License Servers	Manage TS licensing
Windows Authorization Access Group	Access token information
Pre-Windows 2000 Compatible Access	Legacy access

Built-in Global Groups



Group	Purpose
<u>Domain Admins</u>	Full control over the domain
<u>Domain Users</u>	All users in the domain
<u>Domain Guests</u>	Guest access
<u>Domain Computers</u>	All computers in the domain
<u>Domain Controllers</u>	All DCs in the domain
Schema Admins	Modify AD schema (forest root only)
Enterprise Admins	Full control over forest (forest root only)
Group Policy Creator Owners	Create/modify GPOs
RAS and IAS Servers	Access remote access properties
Protected Users	Extra security for privileged accounts
Enterprise Key Admins	Manage key objects (forest root)
Key Admins	Manage key objects (domain)

Built-in Universal Groups



Group	Purpose
<u>Enterprise Admins</u>	Full control over forest (forest root only)
<u>Schema Admins</u>	Modify AD schema (forest root only)
<u>Enterprise Key Admins</u>	Manage key objects (forest root)

Creating Groups

Create a global security group

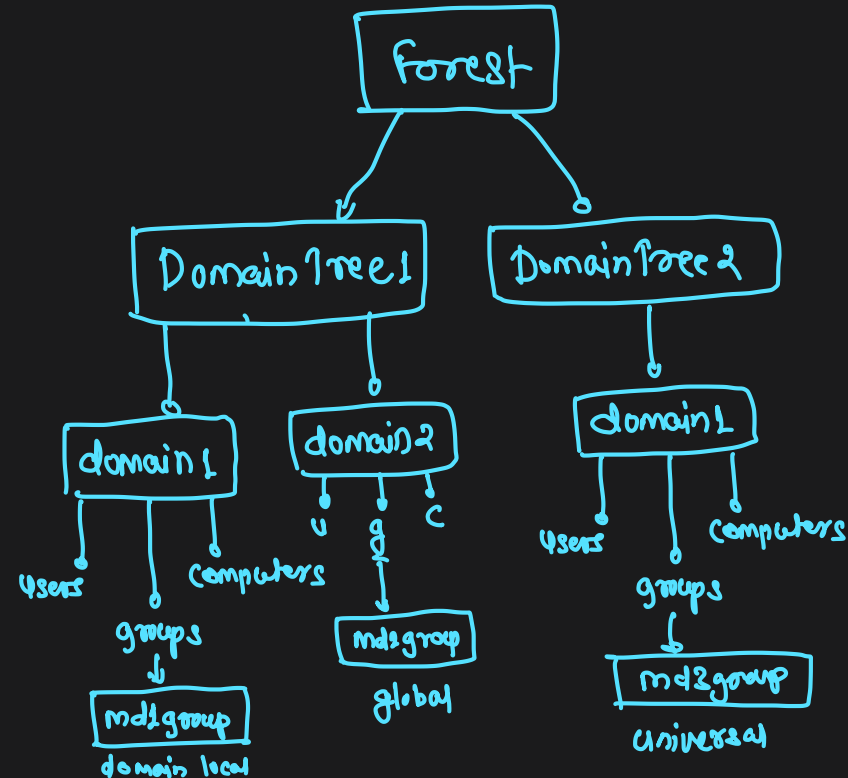
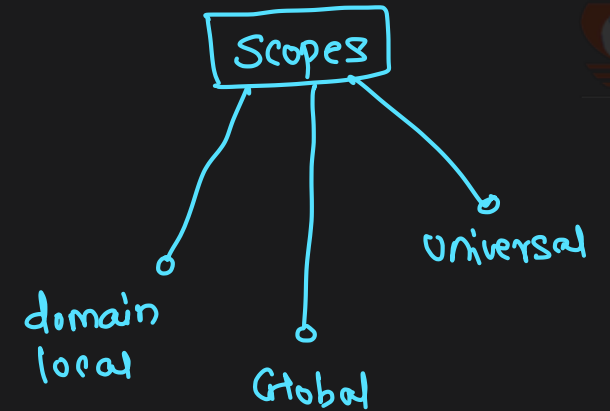
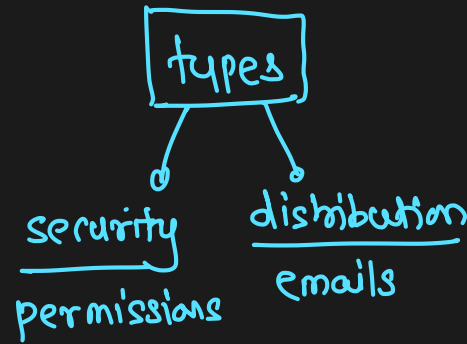
```
New-ADGroup -Name "Sales Team" `
  -SamAccountName "Sales Team" `
  -GroupCategory Security → type
  -GroupScope Global → scope
  -Path "OU=Groups,DC=contoso,DC=com" `
  -Description "All sales department users"
```

Create a domain local security group

```
New-ADGroup -Name "Sales Folder Access" `
  -SamAccountName "Sales Folder Access" `
  -GroupCategory Security `
  -GroupScope DomainLocal `
  -Path "OU=Groups,DC=contoso,DC=com" `
  -Description "Access to Sales shared folder"
```

Create a distribution group

```
New-ADGroup -Name "All Company" `
  -SamAccountName "All Company" `
  -GroupCategory Distribution `
  -GroupScope Universal `
  -Path "OU=Distribution Groups,DC=contoso,DC=com"
```



Managing Groups



Get group info

```
Get-ADGroup -Identity "Sales Team" -Properties *
```

Get group members

```
Get-ADGroupMember -Identity "Sales Team"
```

Get user's groups

```
Get-ADUser -Identity "jdoe" -Properties MemberOf | Select-Object -ExpandProperty MemberOf
```

Add member

```
Add-ADGroupMember -Identity "Sales Team" -Members "jdoe"
```

Add multiple members

```
Add-ADGroupMember -Identity "Sales Team" -Members "jdoe","jsmith","bwilliams"
```

Remove member

```
Remove-ADGroupMember -Identity "Sales Team" -Members "jdoe" -Confirm:$false
```

Remove all members

```
Get-ADGroupMember -Identity "Sales Team" | Remove-ADGroupMember -Identity "Sales Team" -Confirm:$false
```

Managing Groups



```
# Change group scope
Set-ADGroup -Identity "Sales Team" -GroupScope Universal

# Change group type
Set-ADGroup -Identity "Sales Team" -GroupCategory Distribution

# Delete group
Remove-ADGroup -Identity "Sales Team" -Confirm:$false

# Find empty groups
Get-ADGroup -Filter * -Properties Members | Where-Object {$_.Members.Count -eq 0}

# Find groups with no members (alternative)
Get-ADGroup -Filter * | ForEach-Object {
    $members = Get-ADGroupMember -Identity $_ -ErrorAction SilentlyContinue
    if ($members.Count -eq 0) {
        $_.Name
    }
}

# Export group members to CSV
Get-ADGroupMember -Identity "Sales Team" |
    Select-Object Name, SamAccountName, ObjectClass |
    Export-Csv "C:\SalesTeam.csv" -NoTypeInformation
```

Bulk Group Membership



■ CSV Format

- SamAccountName,Group
- jdoe,Sales Team
- jsmith,Sales Team
- bwilliams,Marketing Team

```
Import-Csv "C:\GroupMembership.csv" | ForEach-Object {  
    Add-ADGroupMember -Identity $_.Group -Members $_.SamAccountName  
    Write-Host "Added $($_.SamAccountName) to $($_.Group)"  
}
```

Group Nesting



■ What is Nesting?

- Adding a group as a member of another group
- Allows hierarchical permission assignment

■ Rules

- Global groups can be members of: Domain Local, Universal, Global (same domain)
- Domain Local groups can be members of: Domain Local (same domain)
- Universal groups can be members of: Domain Local, Universal

■ Best Practices

- Avoid deep nesting: Can impact performance and troubleshooting
- Document nesting: Keep a map of group relationships
- Use AGDLP: Standard nesting pattern
- Avoid circular nesting: Can cause issues



User Group Management

- **Add user to a group**

```
Add-ADGroupMember -Identity "SalesTeam" -Members "jdoe"
```

- **Add multiple users at once**

```
Add-ADGroupMember -Identity "SalesTeam" -Members "jdoe", "asmith", "bwayne"
```

- **Add a user to multiple groups at once**

```
Add-ADGroupMember -Identity "SalesTeam", "VPN-Users", "M365-E3" -Members "jdoe"
```

- **Remove a user from a group**

```
Remove-ADGroupMember -Identity "SalesTeam" -Members "jdoe" -Confirm:$false
```

- **Remove multiple users at once**

```
Remove-ADGroupMember -Identity "SalesTeam" -Members "jdoe", "asmith" -Confirm:$false
```



Computers

Computer Object



- A computer object represents a computer joined to the domain
- It Authenticates at startup (machine account)
- It has a password (managed automatically)
- It can be a member of groups
- It can have GPOs applied
- It has a trust relationship with the domain

Creating Computer Objects



■ Via Domain Join

- Computer joins domain → computer object created automatically
- Default location: CN=Computers container
- Can redirect to specific OU via `redircmp` command

■ Via ADUC (Pre-staging)

- Right-click OU → New → Computer
- Enter computer name
- Select "Assign this computer account as a pre-Windows 2000 computer" (optional)
- Click OK

Create computer object

```
New-ADComputer -Name "WS-001" `
    -SamAccountName "WS-001$" `
    -Path "OU=Workstations,DC=contoso,DC=com" `
    -Description "Sales workstation"
```

Pre-stage computer with specific GUID

```
New-ADComputer -Name "WS-002" `
    -SamAccountName "WS-002$" `
    -Path "OU=Workstations,DC=contoso,DC=com" `
    -Description "Marketing workstation"
```

Managing Computer Objects



```
# Get computer info
```

```
Get-ADComputer -Identity "WS-001" -Properties *
```

```
# Find computers by OS
```

```
Get-ADComputer -Filter {OperatingSystem -like "*Windows 11*"} -Properties OperatingSystem
```

```
# Find stale computers (not logged in for 90 days)
```

```
$90DaysAgo = (Get-Date).AddDays(-90)
```

```
Get-ADComputer -Filter {LastLogonTimeStamp -lt $90DaysAgo} -Properties LastLogonTimeStamp
```

```
# Disable computer
```

```
Disable-ADAccount -Identity "WS-001$"
```

```
# Move computer
```

```
Move-ADObject -Identity "CN=WS-001,OU=Workstations,DC=contoso,DC=com" `
    -TargetPath "OU=Stale Computers,DC=contoso,DC=com"
```

```
# Reset computer account
```

```
Reset-ADAccountPassword -Identity "WS-001$" -Reset
```

```
# Delete computer
```

```
Remove-ADComputer -Identity "WS-001" -Confirm:$false
```



Delegation of Administration

- It allows you to grant **specific administrative permissions** to users or groups without making them Domain Admins
- This follows the **principle of least privilege**
- **Delegation Options**
 - **Built-in Groups**
 - **Account Operators:** Manage user/group accounts
 - **Server Operators:** Manage servers
 - **Backup Operators:** Backup/restore
 - **Print Operators:** Manage printers
 - **Delegation of Control Wizard**
 - Right-click OU → Delegate Control
 - Add users/groups to delegate to
 - Select tasks to delegate
 - Create, delete, and manage user accounts
 - Reset user passwords and force password change
 - Or create custom task to delegate
 - Finish